RS70 Yield, Efficiency & Cross-Section Comparison

Existing vs. Latest Results

October 28, 2026

Single Differential Cross-Section Formalism ($\mathrm{d}\sigma/\mathrm{d}p_T$)

LH_2 (pp) Single Differential Cross Section:

$$\frac{\mathrm{d}\sigma_{pp}}{\mathrm{d}p_T} = \frac{1}{\epsilon_{\mathrm{acc}}\mathcal{L}\Delta p_T}\left[\frac{Y^{\mathrm{LH2}}_{\mathrm{total}} - Y^{\mathrm{LH2}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{LH2}}_{\mathrm{signal}}\rangle} - \frac{I_{\mathrm{LH2}}}{I_{\mathrm{flask}}}\left(\frac{Y^{\mathrm{flask}}_{\mathrm{total}} - Y^{\mathrm{flask}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{flask}}_{\mathrm{signal}}\rangle}\right)\right] \tag{1}$$

LD_2 (pd) Single Differential Cross Section:

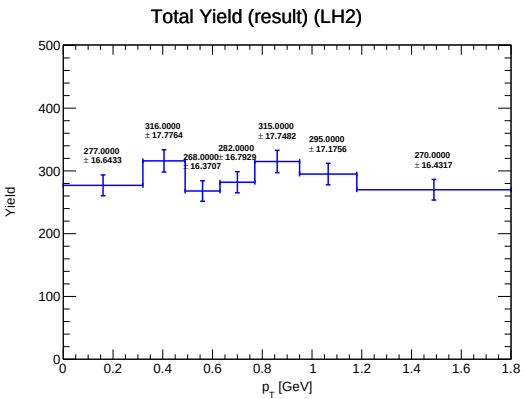
$$\begin{aligned}\frac{\mathrm{d}\sigma_{pd}}{\mathrm{d}p_T} = \frac{1}{\epsilon_{\mathrm{acc}}\mathcal{L}\Delta p_T}\Bigg(&\left[\frac{Y^{\mathrm{LD2}}_{\mathrm{total}} - Y^{\mathrm{LD2}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{LD2}}_{\mathrm{signal}}\rangle} - \frac{I_{\mathrm{LD2}}}{I_{\mathrm{flask}}}\left(\frac{Y^{\mathrm{flask}}_{\mathrm{total}} - Y^{\mathrm{flask}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{flask}}_{\mathrm{signal}}\rangle}\right)\right] \\ &- \frac{T_{HD}}{T_{HH}}\frac{I_{\mathrm{LD2}}}{I_{\mathrm{LH2}}}\left[\frac{Y^{\mathrm{LH2}}_{\mathrm{total}} - Y^{\mathrm{LH2}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{LH2}}_{\mathrm{signal}}\rangle} - \frac{I_{\mathrm{LH2}}}{I_{\mathrm{flask}}}\left(\frac{Y^{\mathrm{flask}}_{\mathrm{total}} - Y^{\mathrm{flask}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{flask}}_{\mathrm{signal}}\rangle}\right)\right]\Bigg)\end{aligned} \tag{2}$$

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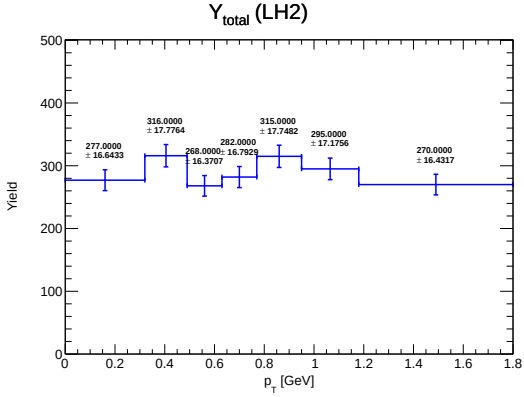
1. Yield Distributions
2. Efficiency Distributions
3. Corrected & Subtracted Yields
4. Cross Sections & Ratios
5. Systematic Uncertainties

LH_2 Target: Total Yield (Y_total_LH2.pdf)

Existing Results

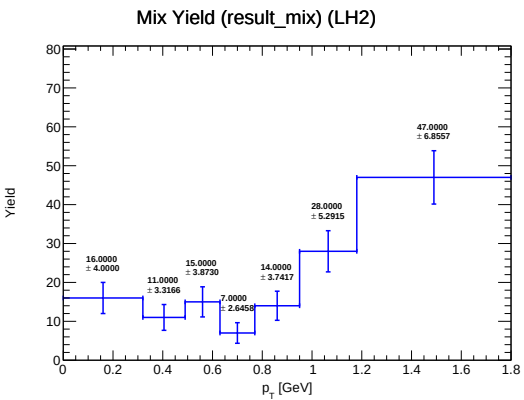


Latest Results

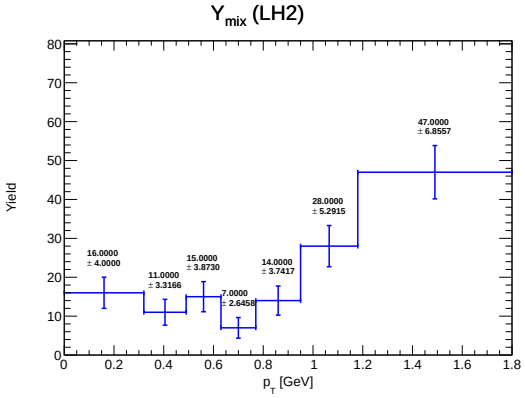


LH_2 Target: Mixed Yield (Y_mix_LH2.pdf)

Existing Results

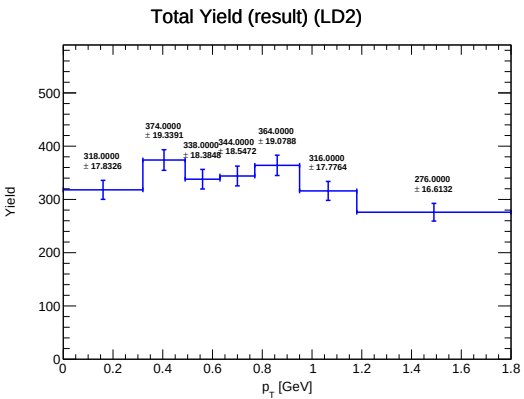


Latest Results

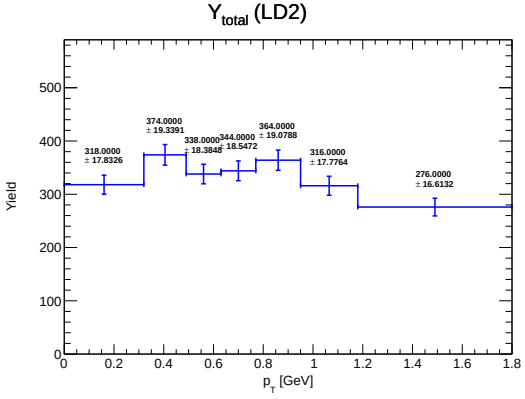


LD_2 Target: Total Yield (Y_total_LD2.pdf)

Existing Results

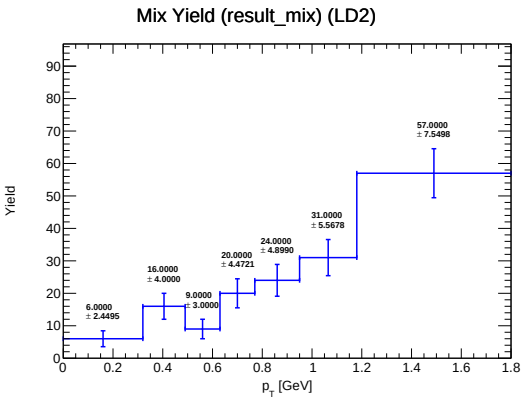


Latest Results

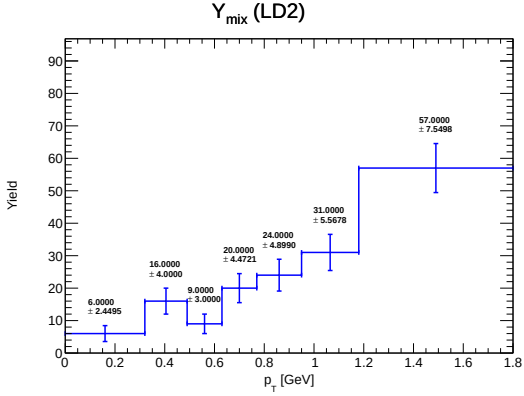


LD_2 Target: Mixed Yield (`Y_mix_LD2.pdf`)

Existing Results

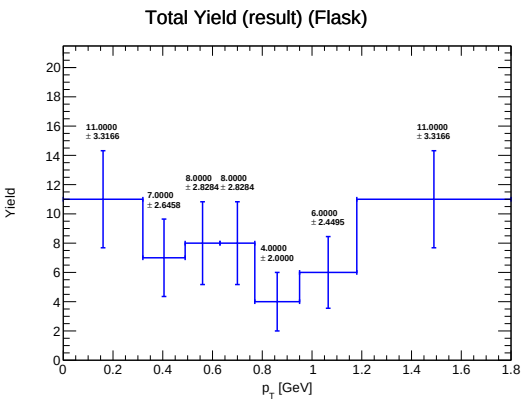


Latest Results

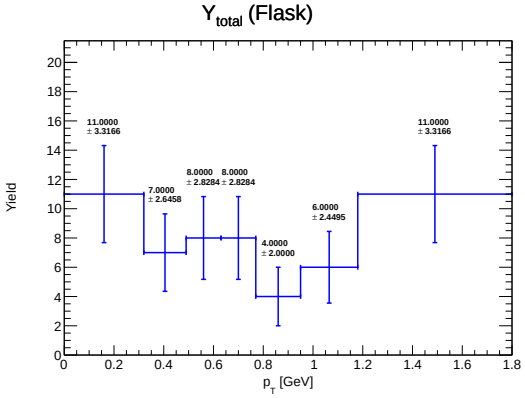


Empty Flask: Total Yield (Y_total_Flask.pdf)

Existing Results

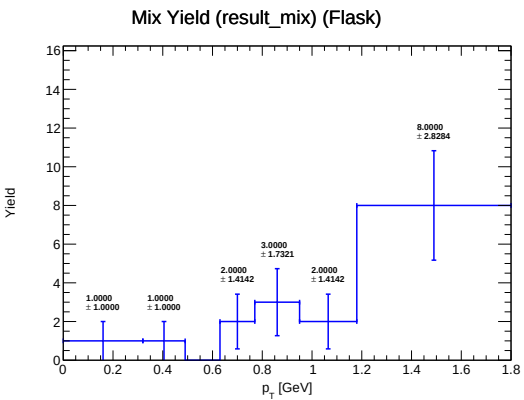


Latest Results

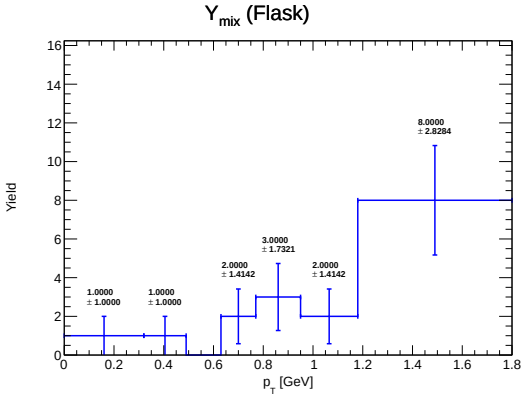


Empty Flask: Mixed Yield (Y_mix_Flask.pdf)

Existing Results

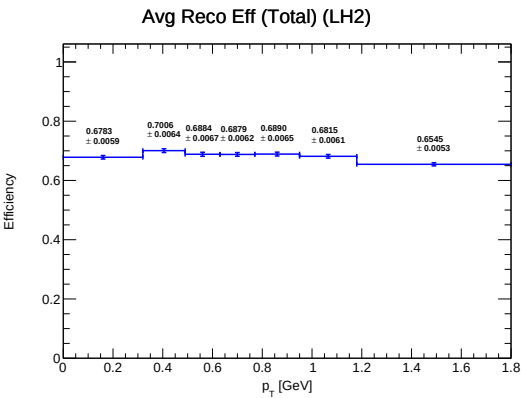


Latest Results

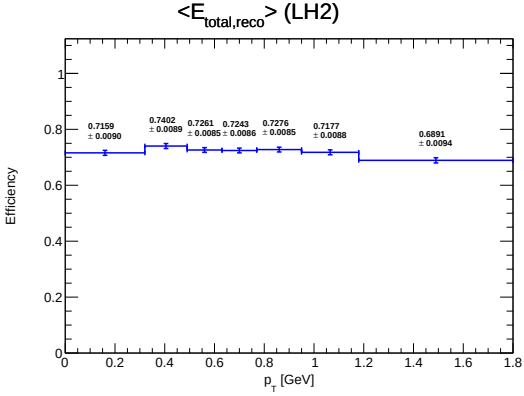


LH_2 Target: Total Reco Efficiency (E_total_reco_LH2.pdf)

Existing Results

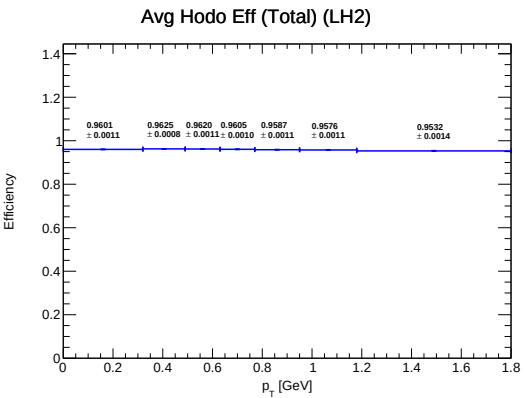


Latest Results

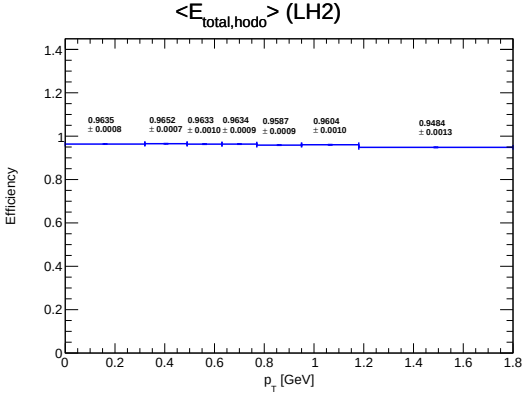


LH_2 Target: Total Hodo Efficiency (E_total_hodo_LH2.pdf)

Existing Results

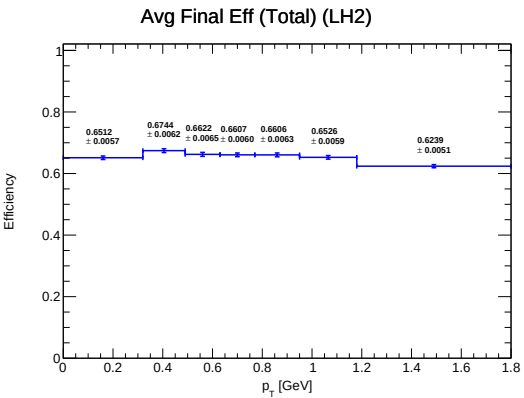


Latest Results

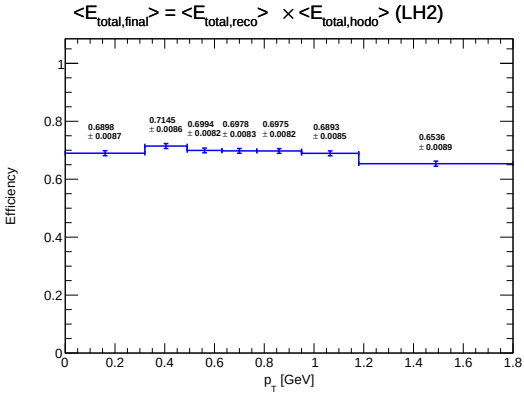


LH_2 Target: Total Final Efficiency (E_total_final_LH2.pdf)

Existing Results

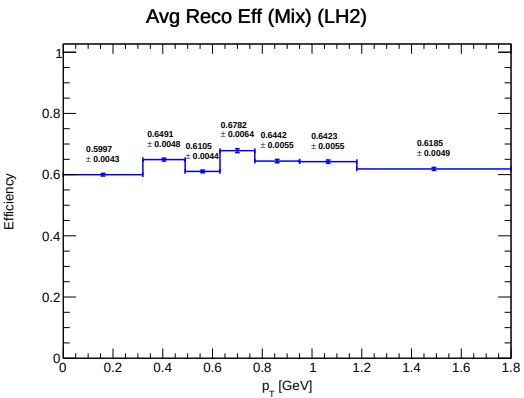


Latest Results

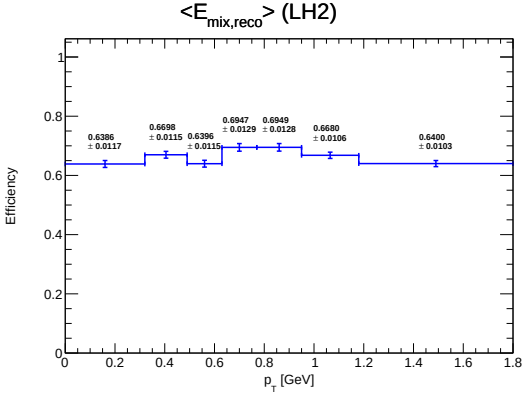


LH_2 Target: Mixed Reco Efficiency (E_mix_reco_LH2.pdf)

Existing Results

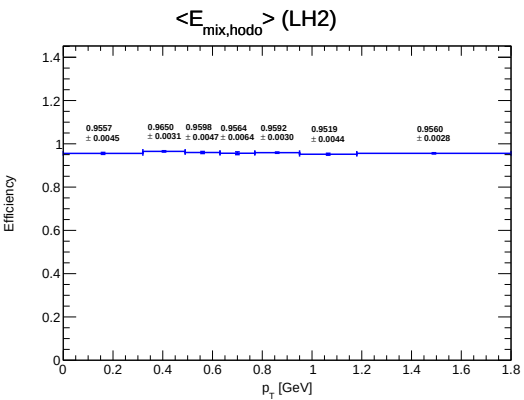


Latest Results

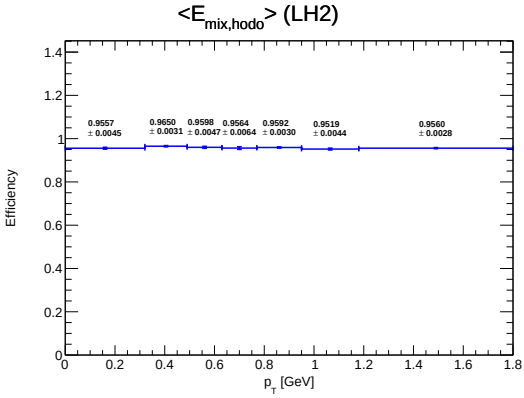


LH_2 Target: Mixed Hodo Efficiency (E_mix_hodo_LH2.pdf)

Existing Results

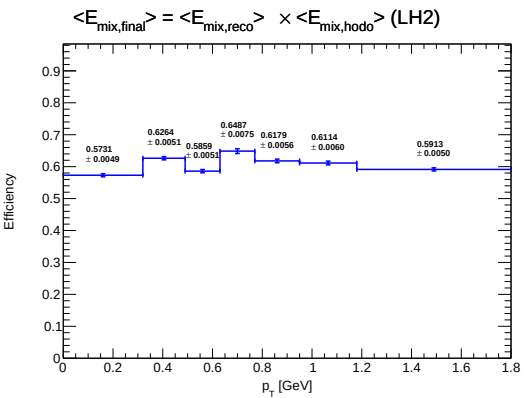


Latest Results

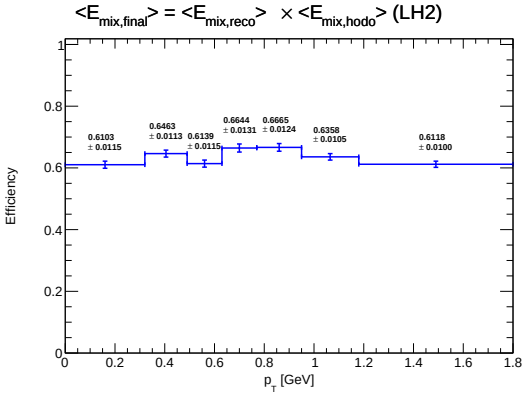


LH_2 Target: Mixed Final Efficiency (E_mix_final_LH2.pdf)

Existing Results

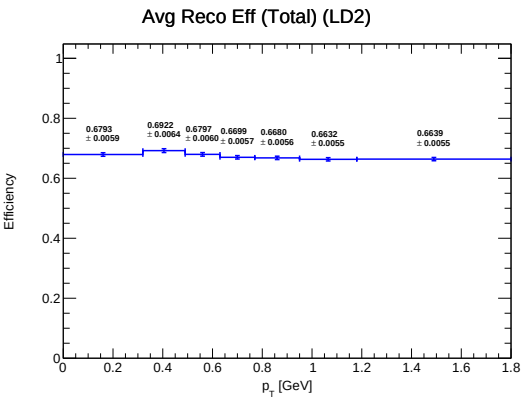


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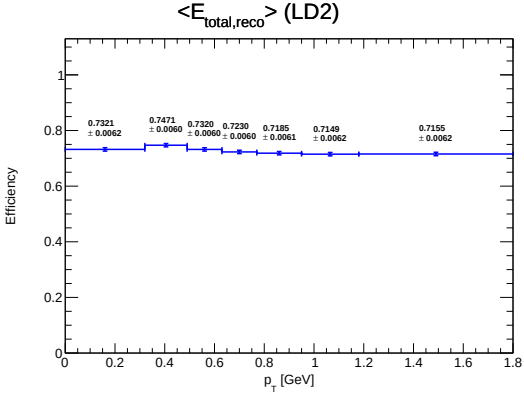


LD_2 Target: Total Reco Efficiency (E_total_reco_LD2.pdf)

Existing Results

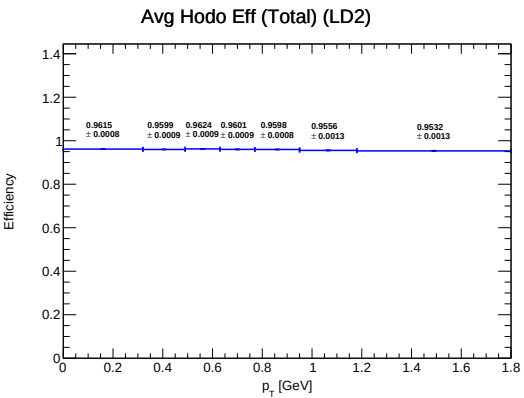


Latest Results

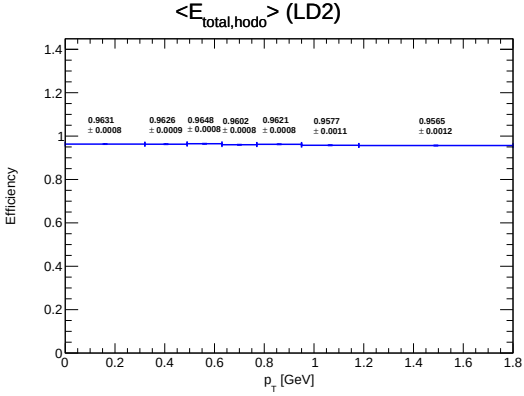


LD_2 Target: Total Hodo Efficiency (E_total_hodo_LD2.pdf)

Existing Results

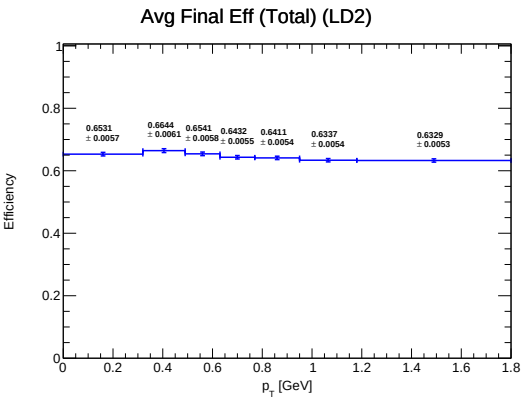


Latest Results

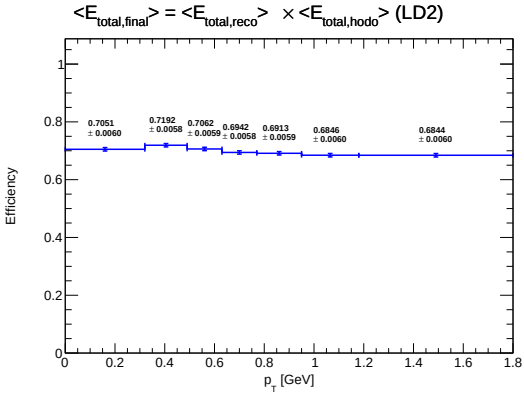


LD_2 Target: Total Final Efficiency (`E_total_final_LD2.pdf`)

Existing Results

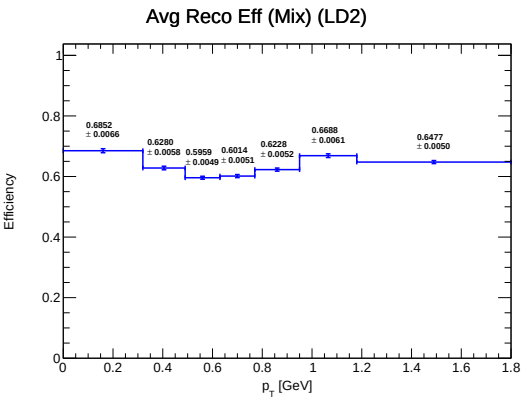


Latest Results

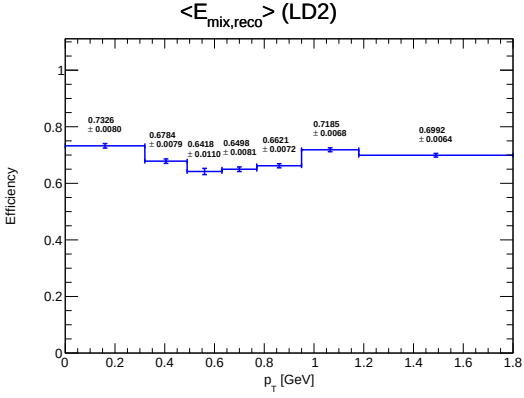


LD_2 Target: Mixed Reco Efficiency (E_mix_reco_LD2.pdf)

Existing Results

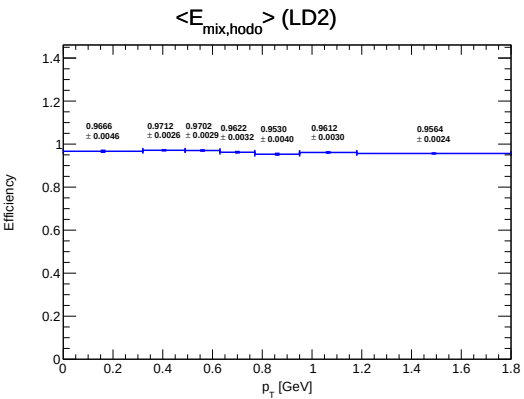


Latest Results

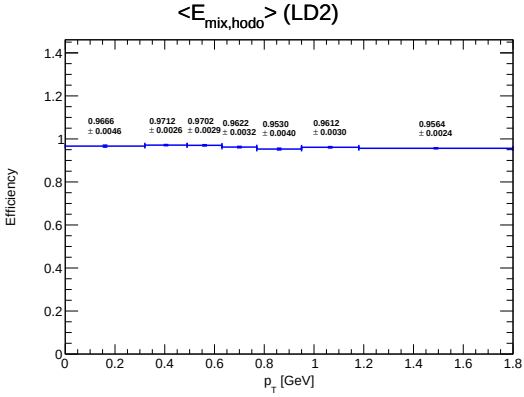


LD_2 Target: Mixed Hodo Efficiency (E_mix_hodo_LD2.pdf)

Existing Results

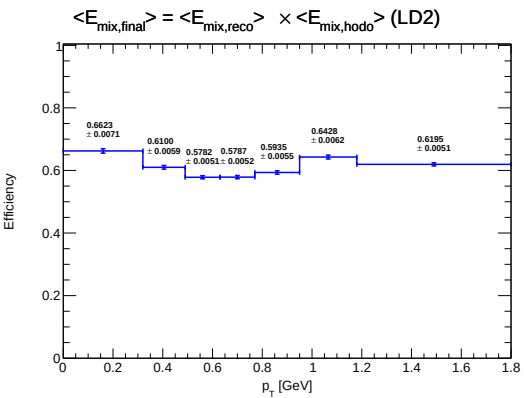


Latest Results

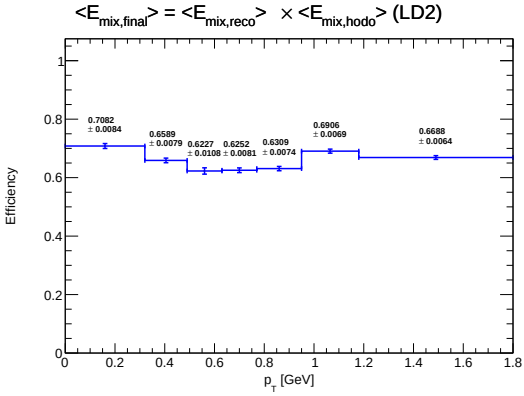


LD_2 Target: Mixed Final Efficiency (E_mix_final_LD2.pdf)

Existing Results

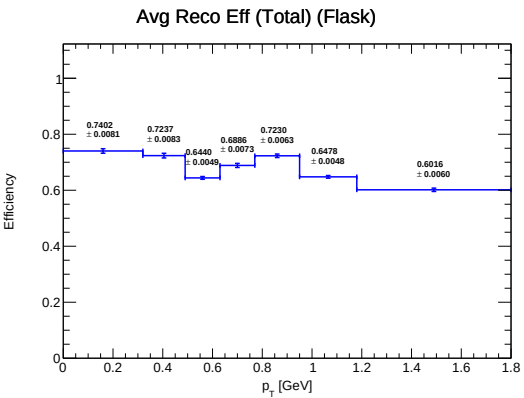


Latest Results

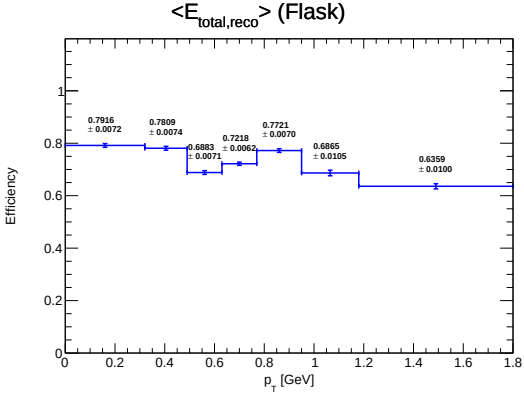


Empty Flask: Total Reco Efficiency (E_total_reco_Flask.pdf)

Existing Results

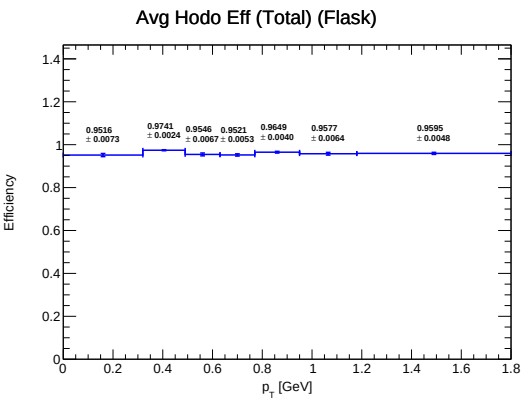


Latest Results

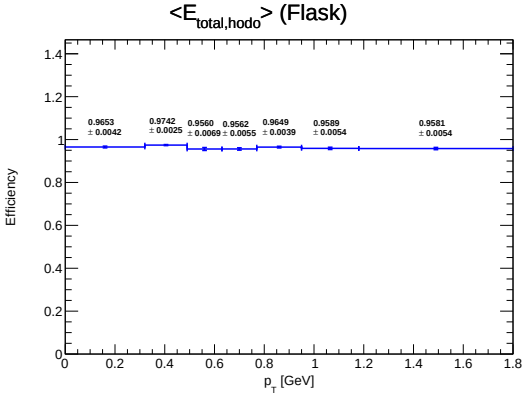


Empty Flask: Total Hodo Efficiency (E_total_hodo_Flask.pdf)

Existing Results

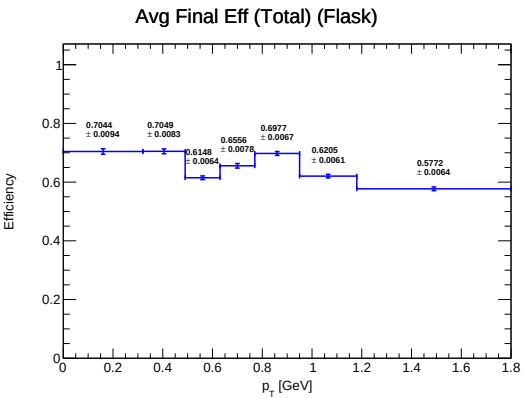


Latest Results

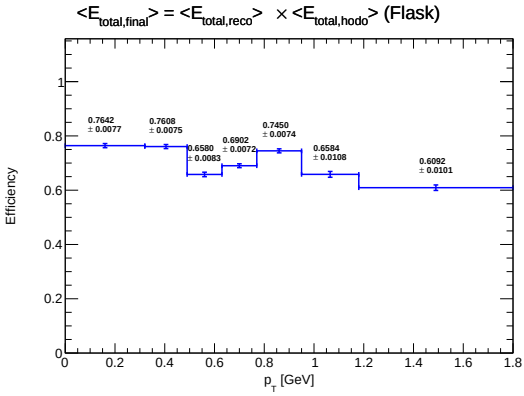


Empty Flask: Total Final Efficiency (E_total_final_Flask.pdf)

Existing Results

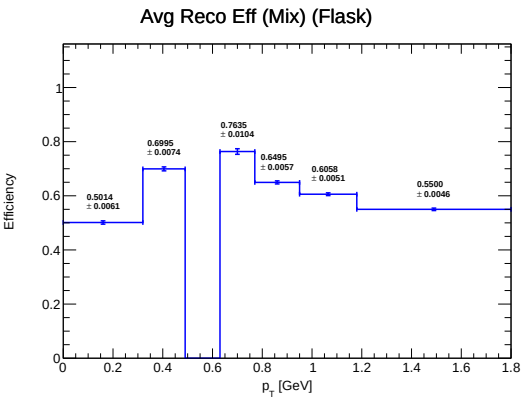


Latest Results

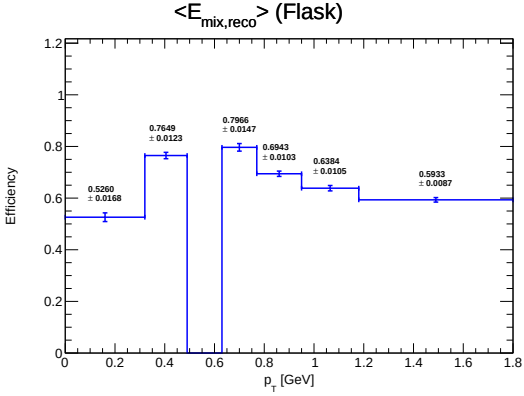


Empty Flask: Mixed Reco Efficiency (E_mix_reco_Flask.pdf)

Existing Results

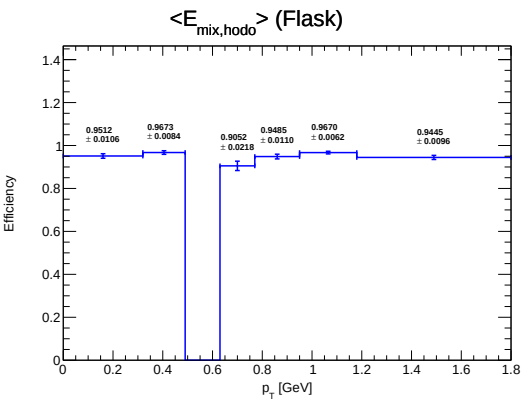


Latest Results

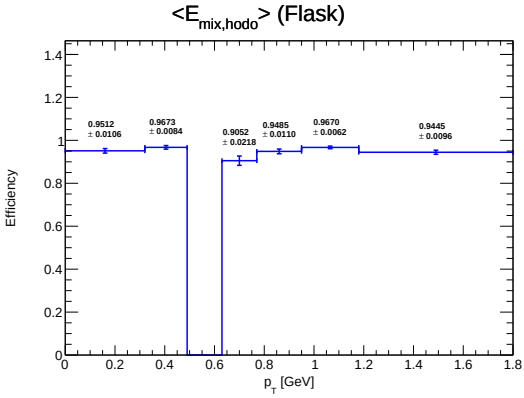


Empty Flask: Mixed Hodo Efficiency (E_mix_hodo_Flask.pdf)

Existing Results

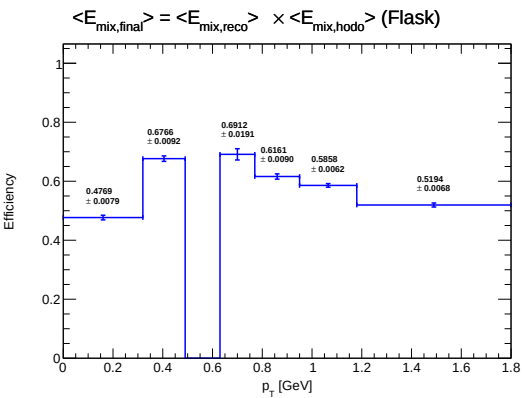


Latest Results

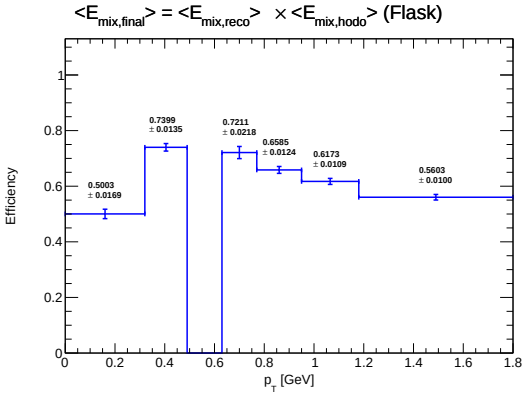


Empty Flask: Mixed Final Efficiency (E_mix_final_Flask.pdf)

Existing Results

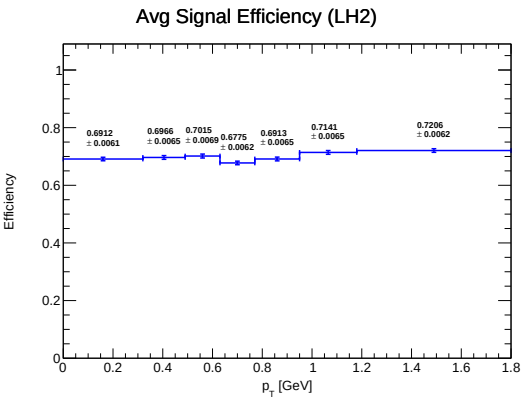


Latest Results

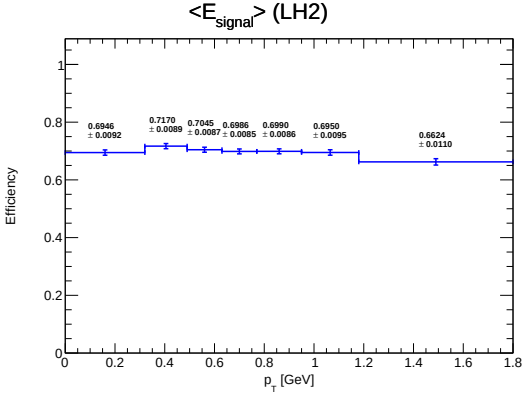


LH_2 Target: Final Signal Efficiency (`E_final_signal_LH2.pdf`)

Existing Results

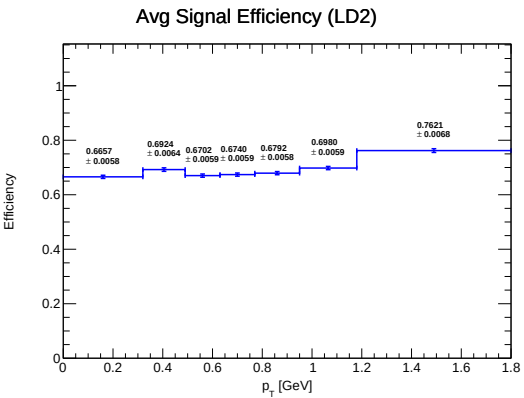


Latest Results

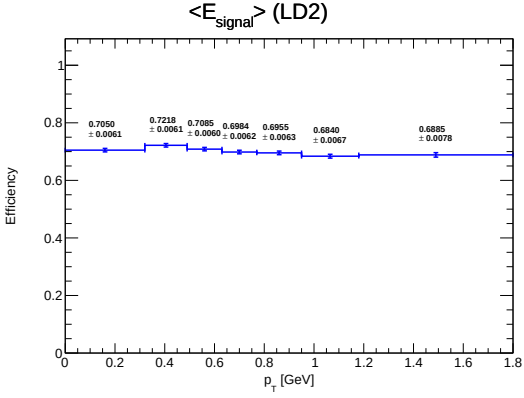


LD_2 Target: Final Signal Efficiency (`E_final_signal_LD2.pdf`)

Existing Results

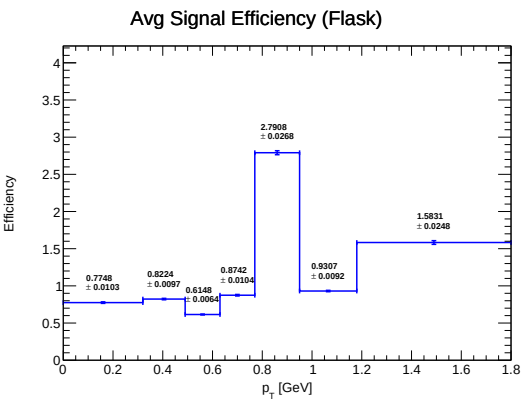


Latest Results

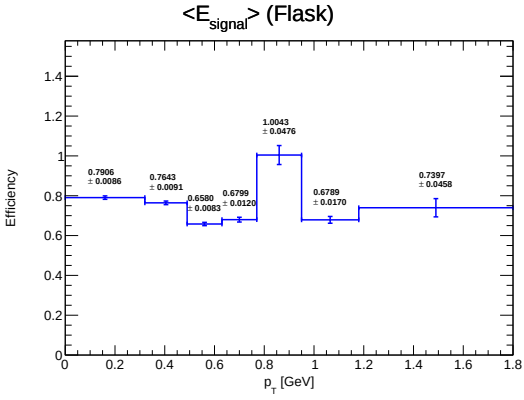


Empty Flask: Final Signal Efficiency (E_final_signal_Flask.pdf)

Existing Results

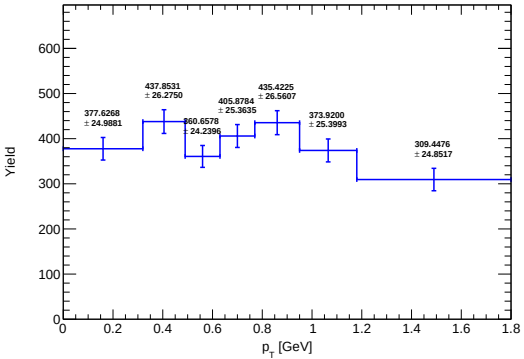


Latest Results

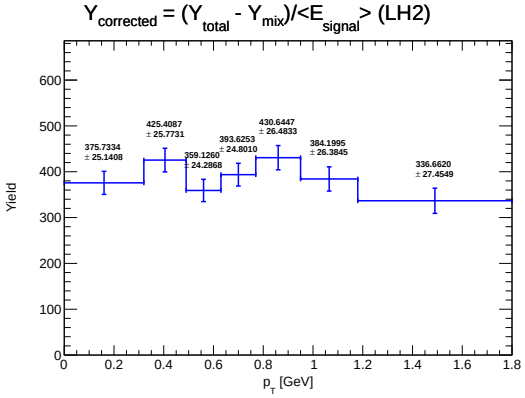


LH_2 Target: Corrected Yield (`Y_corrected_LH2.pdf`)

Existing Results

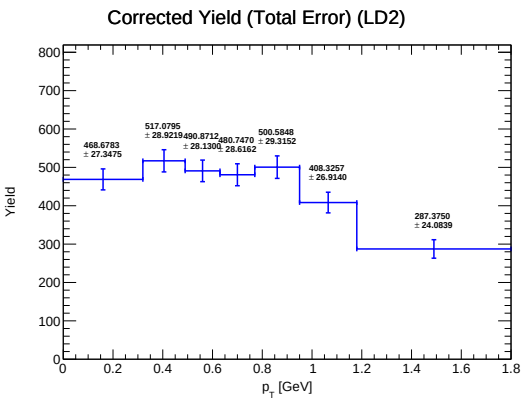


Latest Results

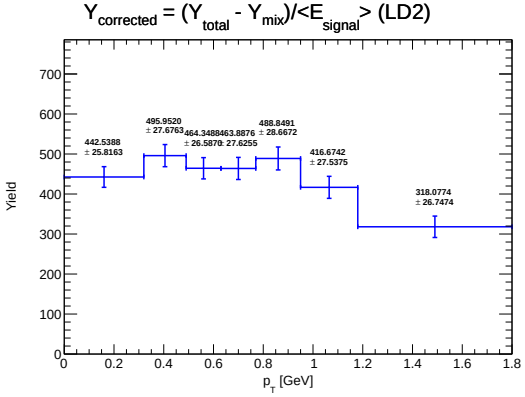


LD_2 Target: Corrected Yield (`Y_corrected_LD2.pdf`)

Existing Results

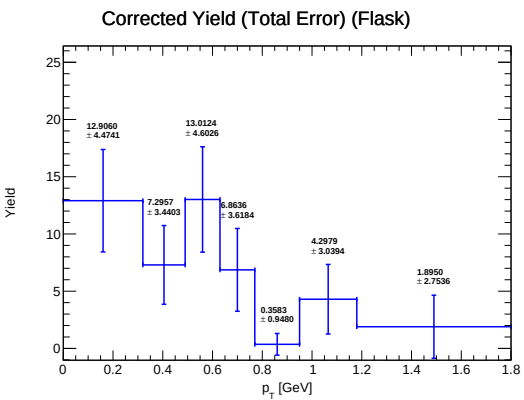


Latest Results

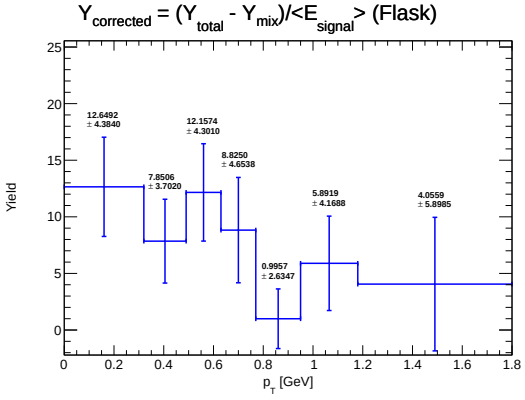


Empty Flask: Corrected Yield (Y_corrected_Flask.pdf)

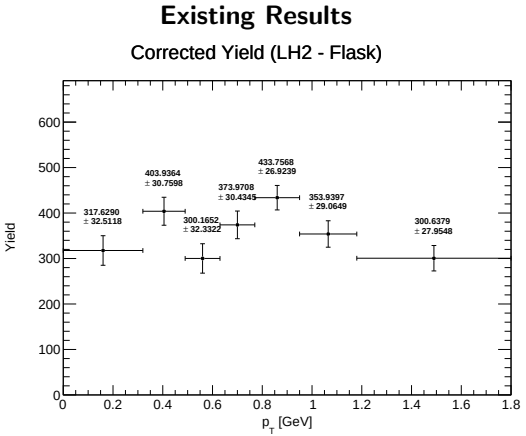
Existing Results

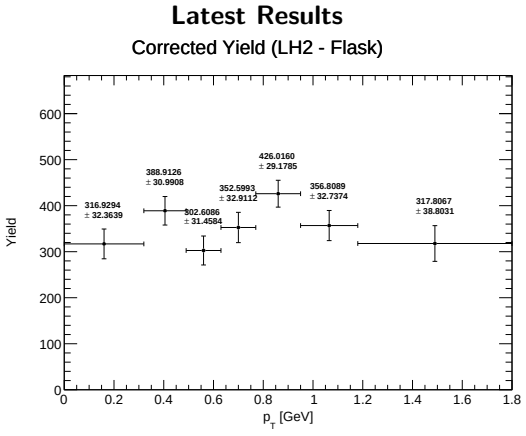


Latest Results



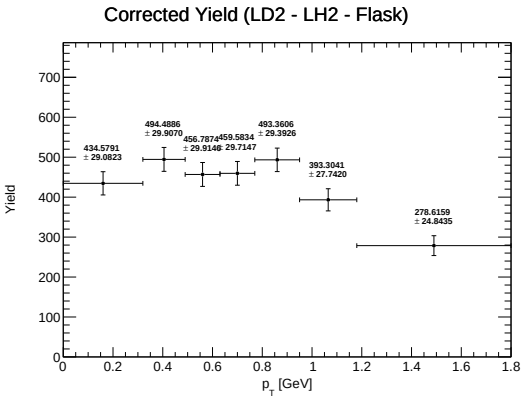
LH_2 Target: Subtracted Corrected Yield (Y_corrected_Subtracted_LH2.pdf)



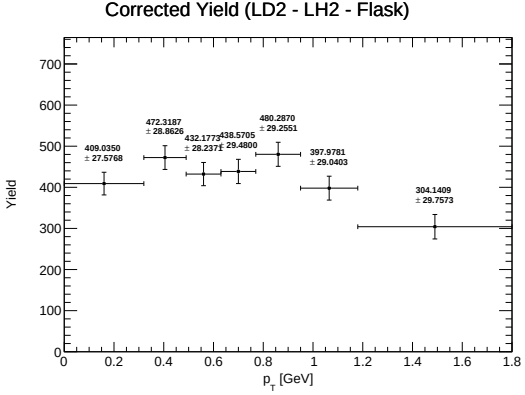


LD_2 Target: Subtracted Corrected Yield (Y_corrected_Subtracted_LD2.pdf)

Existing Results

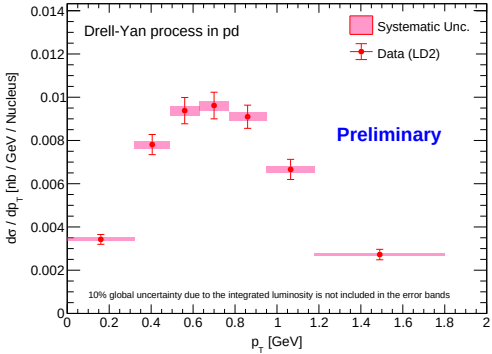


Latest Results

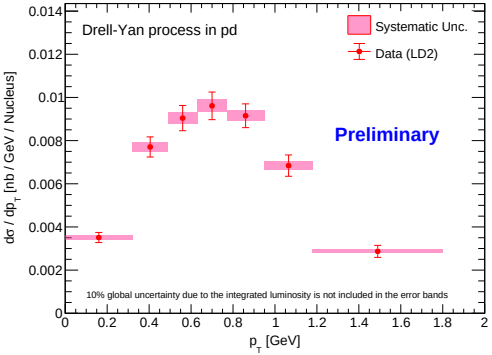


LD_2 Cross Section vs. p_T (CrossSection_LD2_geom_vs_pT.pdf)

Existing Results

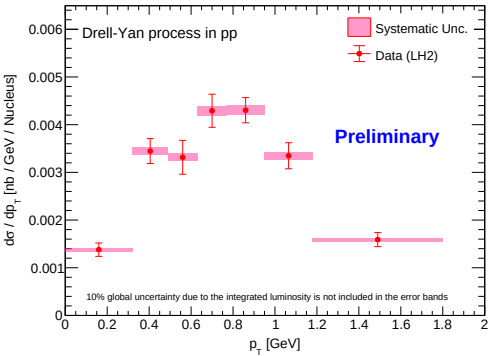


Latest Results

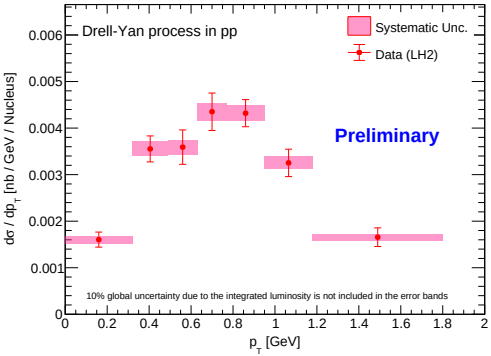


LH_2 Cross Section vs. p_T (CrossSection_LH2_geom_vs_pT.pdf)

Existing Results

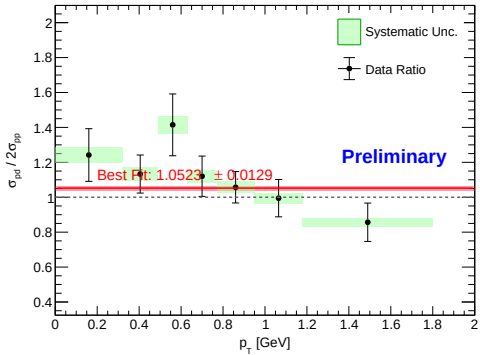


Latest Results

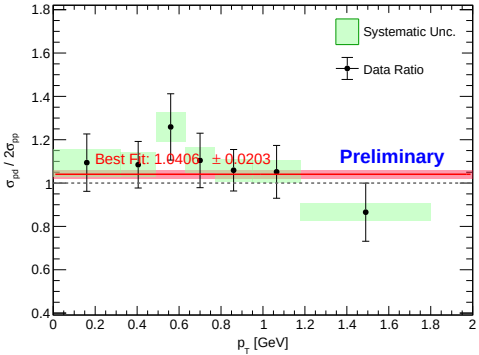


Cross Section Ratio $\sigma_{pd}/2\sigma_{pp}$ vs. p_T (CrossSection_Ratio_pd_2pp_vs_pT_geom.pdf)

Existing Results

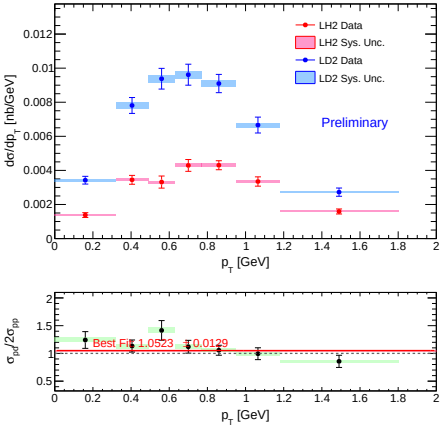


Latest Results

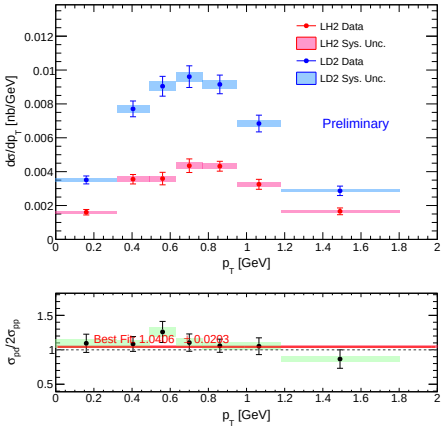


Combined Cross Section & Ratio vs. p_T (Combined_XSec_Ratio_vs_pT_geom.pdf)

Existing Results



Latest Results



LH_2 Target: Systematic Components (Table_Systematics_LH2.tex)

Latest Results

$[p_{T,\min}, p_{T,\max})$	Acc. Corr.	Acc. Err.	Acc. Frac. Err.	ψ' Corr.	ψ' Err.	ψ' Frac. Err.	Yield Corr.	Yield Err.	Yield Frac. Err.
[0.00, 0.32)	0.016654	0.000762	0.045782	0.001829	0.000453	0.247808	316.9294	5.0250	0.015855
[0.32, 0.49)	0.017369	0.000710	0.040850	0.002900	0.000720	0.248166	388.9126	5.3185	0.013675
[0.49, 0.63)	0.016235	0.000683	0.042052	0.003874	0.000957	0.247144	302.6086	4.4877	0.014830
[0.63, 0.77)	0.015614	0.000644	0.041272	0.003034	0.001294	0.426602	352.5993	4.8486	0.013751
[0.77, 0.95)	0.014777	0.000551	0.037271	0.003310	0.000768	0.232102	426.0160	5.2884	0.012414
[0.95, 1.18)	0.012866	0.000475	0.036921	0.004258	0.000933	0.219190	356.8089	5.2742	0.014782
[1.18, 1.80)	0.008346	0.000287	0.034439	0.003692	0.000794	0.215045	317.8067	5.7129	0.017976

Existing Results

$[p_{T,\min}, p_{T,\max})$	Acc. Corr.	Acc. Err.	Acc. Frac. Err.	ψ' Corr.	ψ' Err.	ψ' Frac. Err.	Yield Corr.	Yield Err.	Yield Frac. Err.
[0.00, 0.32)	0.019414	0.000441	0.022716	0.001829	0.000453	0.247808	317.6290	3.4196	0.010766
[0.32, 0.49)	0.018594	0.000392	0.021087	0.002900	0.000720	0.248166	403.9364	4.0769	0.010093
[0.49, 0.63)	0.017444	0.000378	0.021680	0.003874	0.000957	0.247144	300.1652	3.5866	0.011949
[0.63, 0.77)	0.016789	0.000357	0.021237	0.003034	0.001294	0.426602	373.9708	3.7297	0.009973
[0.77, 0.95)	0.015105	0.000297	0.019683	0.003310	0.000768	0.232102	433.7568	4.1235	0.009507
[0.95, 1.18)	0.012398	0.000249	0.020057	0.004258	0.000933	0.219190	353.9397	3.4169	0.009654
[1.18, 1.80)	0.008227	0.000153	0.018561	0.003692	0.000794	0.215045	300.6379	2.6859	0.008934

LD_2 Target: Systematic Components (Table_Systematics_LD2.tex)

Latest Results

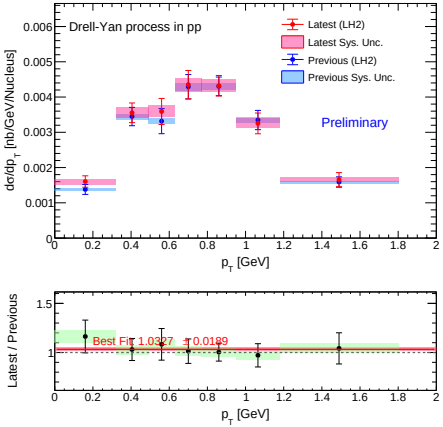
$[p_{T,\min}, p_{T,\max})$	Acc. Corr.	Acc. Err.	Acc. Frac. Err.	ψ' Corr.	ψ' Err.	ψ' Frac. Err.	Yield Corr.	Yield Err.	Yield Frac. Err.
[0.00, 0.32)	0.018640	0.000566	0.030340	0.001829	0.000453	0.247808	409.0350	3.8396	0.009387
[0.32, 0.49)	0.018456	0.000511	0.027677	0.002900	0.000720	0.248166	472.3187	4.1895	0.008870
[0.49, 0.63)	0.017477	0.000496	0.028380	0.003874	0.000957	0.247144	432.1773	3.9687	0.009183
[0.63, 0.77)	0.016688	0.000465	0.027885	0.003034	0.001294	0.426602	438.5705	4.1381	0.009435
[0.77, 0.95)	0.014926	0.000387	0.025908	0.003310	0.000768	0.232102	480.2870	4.4423	0.009249
[0.95, 1.18)	0.012947	0.000333	0.025684	0.004258	0.000933	0.219190	397.9781	4.0889	0.010274
[1.18, 1.80)	0.008756	0.000206	0.023475	0.003692	0.000794	0.215045	304.1409	3.6441	0.011982

Existing Results

$[p_{T,\min}, p_{T,\max})$	Acc. Corr.	Acc. Err.	Acc. Frac. Err.	ψ' Corr.	ψ' Err.	ψ' Frac. Err.	Yield Corr.	Yield Err.	Yield Frac. Err.
[0.00, 0.32)	0.020298	0.000441	0.021727	0.001829	0.000453	0.247808	434.5791	4.1135	0.009465
[0.32, 0.49)	0.019066	0.000392	0.020565	0.002900	0.000720	0.248166	494.4886	4.7868	0.009680
[0.49, 0.63)	0.017807	0.000378	0.021237	0.003874	0.000957	0.247144	456.7874	4.3517	0.009527
[0.63, 0.77)	0.017477	0.000357	0.020401	0.003034	0.001294	0.426602	459.5834	4.1871	0.009111
[0.77, 0.95)	0.015424	0.000297	0.019276	0.003310	0.000768	0.232102	493.3606	4.2804	0.008676
[0.95, 1.18)	0.013143	0.000249	0.018919	0.004258	0.000933	0.219190	393.3041	3.4773	0.008841
[1.18, 1.80)	0.008446	0.000153	0.018081	0.003692	0.000794	0.215045	278.6159	2.5778	0.009252

Cross-Section Comparison vs. p_T: LH_2 and LD_2

LH_2 Plot



LD_2 Plot

